

LECTURE 16

Mapping EER Models to Relational Schemas

Each EER construct, one rule at a time, with its before and after

CSC270 · Database Systems

COMSATS University Islamabad, Lahore Campus

In This Lecture

- 1 The mapping process, applied in a fixed order
- 2 Strong and weak entities, and 1:1, 1:N, M:N relationships
- 3 Multi-valued attributes
- 4 Specialization: three strategies, each worked on the same example
- 5 Categorization, aggregation, and EER constraints

The Mapping Process, in Order

1 Strong entities

2 Weak entities

3 1:1, 1:N, M:N relationships

4 Multi-valued attributes

5 Specialization

6 Categorization and aggregation

7 Constraints

WHY THE ORDER MATTERS

Later rules reuse relations the earlier rules created, so the order is fixed.

Reading the Multiplicity Tells You Where the FK Goes

1:1

Put the FK on the side with mandatory participation.

1:N

Put the FK on the N side, referencing the 1 side.

M:N

Create a new relation. Its key is both foreign keys.

THE TEST

Ask which side can always supply a value, and whether each occurrence repeats. The answer picks the rule.

RULE 1

Strong Entity Types

Before — EER entity

Branch

branchNo

street

city

postcode

After — one relation

Branch(branchNo, street, city, postcode)

branchNo	street	city
B003	163 Main St	Lahore

THE RULE

One relation per strong entity. Its identifier becomes the primary key. Every simple attribute becomes a column.

RULE 2

Weak Entity Types

Before — weak Bed

Bed (weak)

wardNo, bedNo

status

After — composite key

```
CREATE TABLE Bed (  
    wardNo VARCHAR(4) NOT NULL,  
    bedNo SMALLINT NOT NULL,  
    PRIMARY KEY (wardNo, bedNo),  
    FOREIGN KEY (wardNo) REFERENCES Ward  
        ON DELETE CASCADE  
);
```

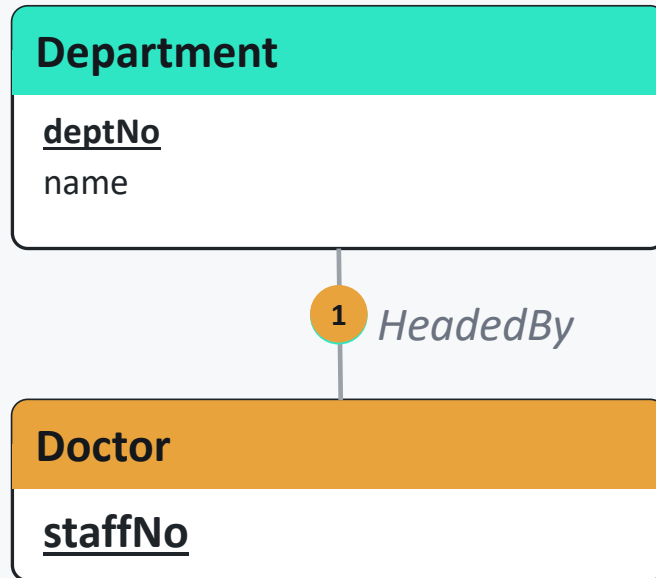
WHY CASCADE

The key is the partial key plus the owner's entire key. ON DELETE CASCADE follows from existence dependency.

RULE 3

1:1 Relationships

Before — Department HeadedBy Doctor



After — FK on Department

Department(deptNo, name, headDoctorNo)

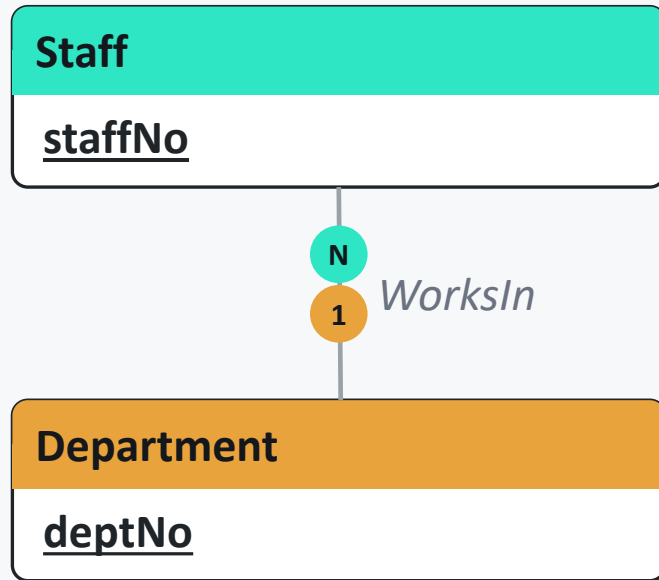
deptNo	name	headDoctorNo
D01	Cardiology	S014

WHICH SIDE

Each department must have a head (total), but not every doctor heads one (partial). The FK goes on the total side.

1:N Relationships

Before — Staff WorksIn Department



After — FK on the many side

`Staff(staffNo, name, deptNo)`

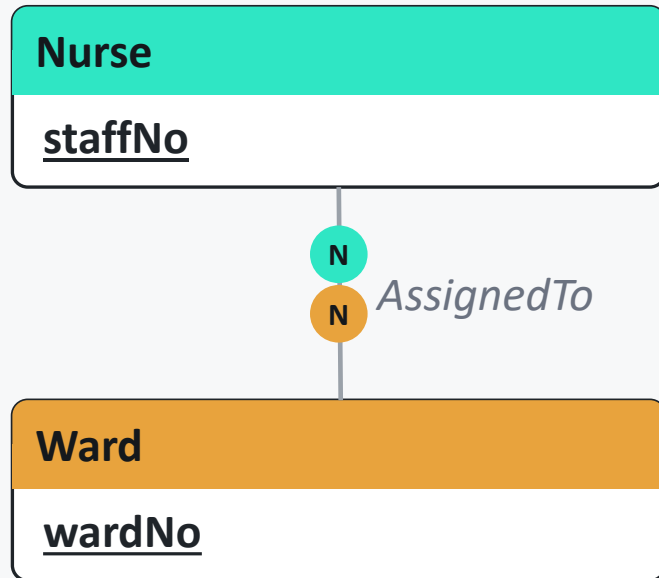
staffNo	name	deptNo
S014	Dr. Amina Raza	D01
S027	Bilal Ahmed	D01

THE RULE

No new relation. Every Staff row carries exactly one deptNo, the “many” side of WorksIn.

M:N Relationships

Before — Nurse AssignedTo Ward



After — a new relation

`NurseWardAssignment(staffNo, wardNo)`

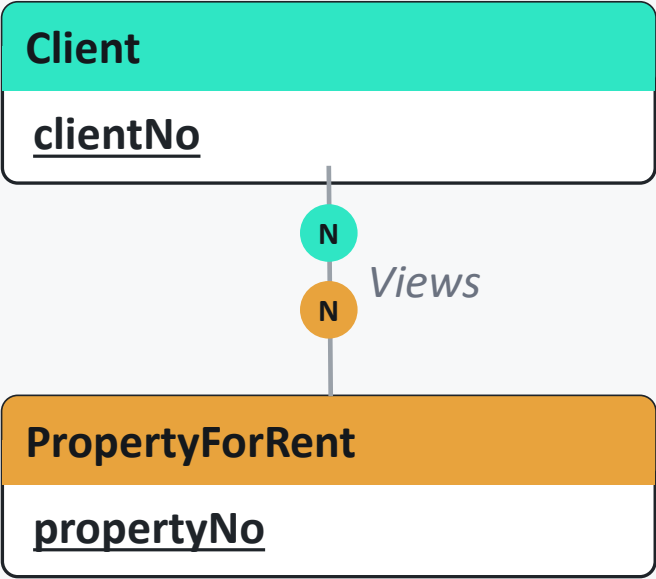
staffNo	wardNo
S027	W1
S027	W2
S041	W1

THE RULE

Neither key alone identifies one assignment. The primary key is both foreign keys together.

A Relationship With Its Own Attributes

Before — Client Views PropertyForRent



After — Viewing relation

`Viewing(clientNo, propertyNo, viewDate, comment)`

clientNo	propertyNo	viewDate
C12	PA14	2026-02-03

THE SAME RULE

viewDate and comment belong to the pairing. The M:N rule applies, and the relation's key is both foreign keys.

RULE 6

Multi-Valued Attributes

Before — Patient phoneNumbers {}

Patient

patientNo

phoneNumbers {}

After — its own relation

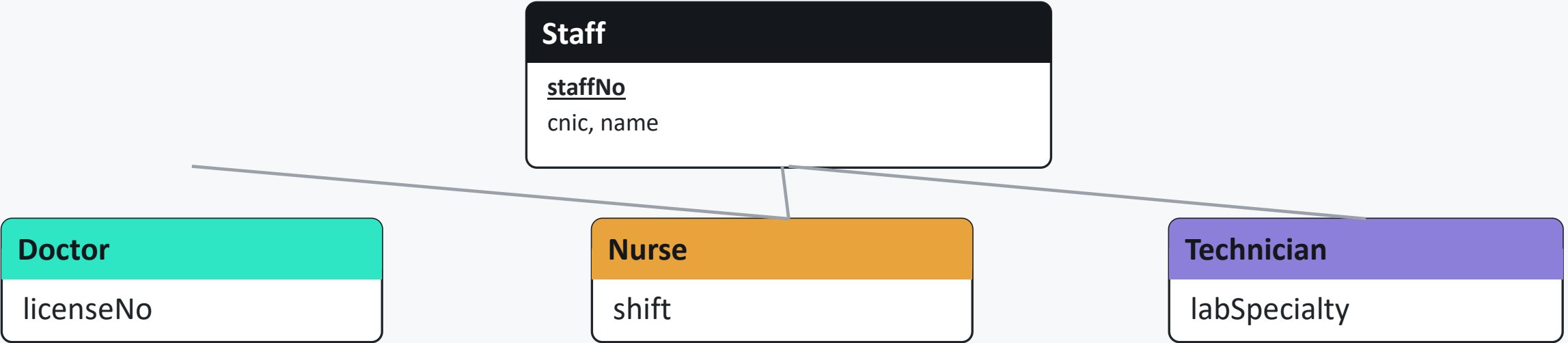
PatientPhone(patientNo, phoneNumber)

patientNo	phoneNumber
P0142	0300-1234567
P0142	021-34567890
P0198	0333-9876543

THE RULE

A single column holds one value, so a multi-valued attribute always becomes its own relation.

Three Strategies, One Example



Strategy	Works when	Trade-off
A subclass-only	Disjoint and total	Duplicated columns; UNION for all staff
B superclass + subclasses	Any constraints	One join to rebuild a row
C single table + type	Disjoint, small subclasses	Many NULLs

Subclass-Only: One Relation per Leaf

Staff is disjoint and total, so the superclass row is never needed on its own.

`Doctor(staffNo, cnic, name, licenseNo, fee)`

staffNo	cnic	name	licenseNo	fee
S014	35202-...	Dr. Amina	PMC-4471	2500

`Nurse(staffNo, cnic, name, shift)`

staffNo	cnic	name	shift
S027	37405-...	Bilal Ahmed	Night

THE COST

Shared columns (cnic, name) are copied into every leaf. “All staff” needs a UNION across the leaves.

Single Table With a Discriminator

Every subclass column folds into one table. staffType says which subclass a row is.

`Staff(staffNo, name, staffType, licenseNo, shift, labSpecialty)`

staffNo	name	staffType	licenseNo	shift	labSpecialty
S014	Dr. Amina	Doctor	PMC-4471	NULL	NULL
S027	Bilal	Nurse	NULL	Night	NULL

```
ALTER TABLE Staff ADD CONSTRAINT chk_staff_type
CHECK (staffType IN ('Doctor','Nurse','Technician','Administrator'));
```

THE COST

NULLs grow with each subclass: the attribute overload from Lecture 14, back in the schema.

Superclass + Subclasses for the Hospital

Staff(staffNo, cnic, name, dateJoined, deptNo)

staffNo	cnic	name
S014	35202-...	Dr. Amina Raza

Doctor(staffNo, licenseNo, consultationFee)

staffNo	licenseNo	consultationFee
S014	PMC-4471	2500

Consultant(staffNo, yearsOfExperience, officeRoomNo)

staffNo	yearsOfExperience	officeRoomNo
S014	12	C-204

WHY STRATEGY B FOR THE HOSPITAL

Resident.supervisingConsultantNo points into Consultant.staffNo, not Staff, because only consultants supervise.
Strategy B composes across both levels.

RULE 8

Categorization (Union Types)

Before — RegisteredOwner

RegisteredOwner
<u>ownerNo</u>
ownerSince

After — surrogate key

`RegisteredOwner(ownerNo, ownerSince)`

ownerNo	ownerSince
RO-001	2019-03-01

`Person(cnic, name, ownerNo)`

cnic	name	ownerNo
35202-...	Hina Tariq	RO-001

THE RULE

Person, Bank, and Company have unrelated keys, so the category gets a surrogate key. Each owner type gets a nullable FK to it.

Aggregation Leaves No Special Trace

```
CREATE TABLE Admission (  
    admissionNo    VARCHAR(8) PRIMARY KEY,  
    admissionDate  DATE NOT NULL,  
    patientNo      VARCHAR(6) NOT NULL REFERENCES Patient,  
    wardNo         VARCHAR(4) NOT NULL,  
    bedNo          SMALLINT NOT NULL,  
    attendingDoctorNo VARCHAR(6) NOT NULL REFERENCES Doctor,  
    FOREIGN KEY (wardNo, bedNo) REFERENCES Bed  
);
```

THE RULE

The aggregate maps like any entity with attributes. Its outer relationship, AttendedBy, becomes one ordinary foreign key.

KEY POINT

Aggregation is a modeling tool. Nothing special survives into the DDL.

Mandatory Participation and CHECK

Before — mandatory (1,1)

PropertyForRent

propertyNo

branchNo (1,1)

After — NOT NULL

```
branchNo VARCHAR(4) NOT NULL
```

Disjoint, under Strategy C: a CHECK on the discriminator

```
CHECK (staffType IN ('Doctor','Nurse','Technician','Administrator'))
```

WHAT SURVIVES

Mandatory participation maps to NOT NULL. Disjointness maps to CHECK only under Strategy C.

When a Foreign Key Cannot Enforce (1,*)

- The rule: every Department has at least one Staff member.
- A foreign key stops a Staff row pointing nowhere, and stops one row pointing at two departments.
- It cannot require that some Staff row points at every Department.

A REAL, COMMON GAP

Insert a Department with no staff: the database accepts it. Enforce (1,*) with a trigger or an application check.

Every EER Construct, Mapped

EER construct	Relational mapping
Strong entity	One relation; identifier is the primary key
Weak entity	Partial key + owner's entire key
1:1	FK on the mandatory-participation side
1:N	FK on the many side
M:N	New relation; key is both foreign keys
Multi-valued attribute	New relation; owner's key + attribute
Specialization	Strategy A, B, or C
Aggregation	Aggregate maps as usual; outer link is one FK
Composition	1:N plus ON DELETE CASCADE
Categorization	Surrogate key; nullable FK per superclass

KEY TAKEAWAYS

Key Takeaways

- 1 Mapping runs in a fixed order, because later rules reuse relations the earlier rules created.
- 2 1:1 and 1:N need only correct FK placement; M:N and multi-valued attributes always need a relation.
- 3 Strategy B is the safe default, and the only one that composes across multi-level hierarchies.
- 4 Some constraints survive to SQL only partially: a true (1,*) needs a trigger or application check.